

CS228 : Logic for Computer Science

Module 1: Propositional Logic

Lecture 3: Formal Proofs and Natural Deduction

Instructor: S. Akshay

IIT Bombay, India

Recap

Propositional Logic

- ▶ **Syntax**
 - ▶ Atomic propositions
 - ▶ Propositional formula
 - ▶ Parse trees
- ▶ **Semantics**
 - ▶ Truth tables

Examples of puzzles encoded into Propositional logic.

Problems of interest

- ▶ **Satisfiability** and **validity**
 - ▶ Does there exist a model/satisfying assignment?
- ▶ **Semantic entailment** and **equivalence**

Semantic Entailment and Equivalence (Recap)

Definition: Consider propositional formulas F, G over **AP**.

Whenever F evaluates to 1, G also evaluates to 1 — we say, $F \models G$.

That is, for all assignments m of **AP**, if $m(F) = 1$ then $m(G) = 1$.

Definition (Semantic Equivalence)

We say that F and G are **semantically equivalent**, denoted $F \equiv G$, if $F \models G$ and $G \models F$.

That is, for all assignments m of **AP**, $m(F) = 1$ if and only if $m(G) = 1$.

Example: $(p \rightarrow q) \equiv (\neg p \vee q)$.

- ▶ How do we check semantic entailment and equivalence using truth tables?
 - ▶ If two formulas have same truth table, then they are equivalent!
 - ▶ What about the converse? Are there two equivalent formulas that have different truth tables?

Semantic equivalence can be used to simplify formulas

- ▶ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ (Associativity)
- ▶ $(p \oplus q) \oplus r \equiv p \oplus (q \oplus r)$ (Associativity)
- ▶ $(p \wedge q) \vee r \equiv (p \vee r) \wedge (q \vee r)$ (Distributivity)

p	q	r	$p \wedge q$	$(p \wedge q) \wedge r$	$q \wedge r$	$p \wedge (q \wedge r)$
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	1	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	1	0	0	0
1	1	1	1	1	1	1

Same in every row $\Rightarrow$ the two formulas are semantically equivalent!

Pop Quiz!

Question 1 (1 mark)

Let p = “you attend the lectures”, q = “you fall asleep”, r = “you will pass the course”.

Write a propositional formula for: “If you attend the lectures, and you don’t fall asleep, then you will pass the course.”

Question 2 (2 marks)

Show, using truth tables, that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ (Distributivity).

Question 3 (3 marks)

Consider the formula $\varphi = (p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$.

- (i) Construct its truth table.
- (ii) Is φ valid, i.e., does it evaluate to true under all assignments?
- (iii) Is $\neg\varphi$ satisfiable and/or valid?

An aside: Minimal logical connectives

We used 2 0-ary, 1 unary, and 5 binary connectives to describe the propositional logic.

However, it is not the minimal set needed for the maximum expressivity.

Exercise 2.8

Show that $\neg$ and $\vee$ can define the whole propositional logic. That is, show the following:

1. $\top \equiv p \vee \neg p$ for some $p \in \mathbf{AP}$
2. $\perp \equiv \neg \top$
3. $(p \wedge q) \equiv \neg(\neg p \vee \neg q)$
4. $(p \oplus q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$
5. $(p \rightarrow q) \equiv (\neg p \vee q)$
6. $(p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

Similarly, show that $\neg$ and $\wedge$ can define all the other connectives.

Exercise 2.9

A **universal connective** is a logical connective, that can alone define all of propositional logic.

1. Give an example of a universal connective. Are there several universal connectives?
2. * Show $\oplus$ is not a universal connective!

How can we show that $\oplus$ is not universal?

We will show that $\oplus$ alone cannot define $\neg$. We start with a property about $\oplus$ -only formulas.

Lemma

Any formula φ built only from $\oplus$ and propositional variables (no constants) evaluates to **false** whenever **all** its variables are false.

Proof (by structural induction on the formula)

- ▶ **Base case:** $\varphi = p$. Under the all-false valuation, φ evaluates to 0.
- ▶ **Inductive step:** $\varphi = \varphi_1 \oplus \varphi_2$. By IH, both φ_1 and φ_2 evaluate to 0 under the all-false valuation, so $\varphi_1 \oplus \varphi_2 = 0 \oplus 0 = 0$.

Now, we argue by contradiction.

Suppose some $\oplus$ -only formula $\varphi(p, q_1, \dots, q_n)$ satisfied $\varphi \equiv \neg p$. Set p and all q_i to false: by Lemma, φ evaluates to 0, but $\neg p$ evaluates to 1 when $p = 0$. **Contradiction!** □

Other problems of interest

- ▶ Uniqueness: Does a given formula have a unique satisfying assignment?
- ▶ Counting: How many satisfying assignments does the formula have?
 - ▶ Decision Variant: Given Formula F and number k , is the number of satisfying assignments of F more than k ?

All these can be answered from the truth table

So what's the big problem?

The problem with truth tables

- ▶ If the formula has 100 variables (typically encodings that are useful in practice require this!)
- ▶ Then, number of lines of truth table is 2^{100} ($\sim 10^{30}$).
- ▶ More than all the digital storage on Earth!
- ▶ So, this approach doesn't work for anything except toy examples...

So what can we do?

Solution:

1. Formal proof systems
2. Knowledge compilation

Topic 3.1

A Formal Proof system

How do we reason with formulas?

Reasoning about formulas via Truth Tables is not always practical

But do we really need the full Truth Table always?

- ▶ Is $((p \vee q) \rightarrow (q \wedge r)) \wedge (\neg((p \vee q) \rightarrow (q \wedge r)))$ **satisfiable**?
- ▶ Does $p \wedge q \models p$ **hold**?
- ▶ If q is true then $p \vee q$ is **also true**.
- ▶ If I have my umbrella open, and I only open it when it's raining, then **it is raining**.

For all these questions, do we need to write Truth tables?

We use **rules** between propositions and connectives!

Can we formalize them?

A proof system

Our goal is to define rules so that we can write a proof!

Definition: Sequent

Suppose we have

- ▶ Formulas $F_1, \dots, F_n$: call them **premises**
- ▶ Another formula G : call it **conclusion**
- ▶ We write

$$F_1, \dots, F_n \vdash G$$

to express the *claim* that, starting from $F_1, \dots, F_n$, by applying the “rules” repeatedly, one can derive G .

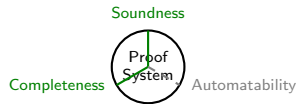
- ▶ This expression is called a **sequent**.
- ▶ The sequent is said to be **valid** if such a derivation actually exists, i.e., you can actually derive G from $F_1, \dots, F_n$ using the rules.

A proof system

What are our guiding principles to define rules?

E.g., we want to obtain $F, F \rightarrow G \vdash G$, but not $p, q \vdash p \wedge \neg q$.

- ▶ **Soundness**: whatever we prove is “correct”:
 - ▶ If $F_1, \dots, F_n \vdash G$ is valid, then $F_1 \dots F_n \models G$ holds.
- ▶ **Completeness**: whatever is correct can be proved
 - ▶ If $F_1, \dots, F_n \models G$ holds, then $F_1 \dots F_n \vdash G$ is valid.
- ▶ **Automatability**



(a logician might not mind losing this 3rd spoke!)

(General) Semantic Entailment

$$F_1, \dots, F_n \models G$$

if for all valuations in which $F_1, \dots, F_n$ evaluate to 1, G also evaluates to 1.

(Recall: a **valuation** is just an assignment of a truth value to each atomic proposition.)

A formal proof system: Natural deduction

Natural deduction rules (from Huth & Ryan's Book)

12 Axioms or rules:

1. Introducing and Eliminating $\wedge$
2. Introducing and Eliminating $\rightarrow$
3. Introducing and Eliminating $\vee$
4. Negation rules: Introducing, Eliminating and Double negation
5. Bot rule

4 Derived/inferred rules to make life easier!



Gerhard Gentzen (1909-1945)

Conjunction Rules

If my cat is brown and it is fat then my cat is brown and fat!

$$\frac{F \quad G}{F \wedge G} \wedge i$$

$$\frac{F \wedge G}{F} \wedge e_1$$

$$\frac{F \wedge G}{G} \wedge e_2$$

- ▶ Rules to Introduce and Elimination $\wedge$
- ▶ **Exercise 3.1 Show that $p \wedge q, r \vdash q \wedge r$ is a valid sequent.**

1. $p \wedge q$ premise
2. r premise
3. q $\wedge e_1$ 1
4. $q \wedge r$ $\wedge i$ 3,2

Conjunction Rules

If my cat is brown and it is fat then my cat is brown and fat!

$$\frac{F \quad G}{F \wedge G} \wedge i$$

$$\frac{F \wedge G}{F} \wedge e$$

$$\frac{F \wedge G}{G} \wedge e$$

- ▶ Rules to Introduce and Elimination $\wedge$
- ▶ **Exercise 3.1 Show that $p \wedge q, r \vdash q \wedge r$ is a valid sequent.**

1. $p \wedge q$ premise
2. r premise
3. q $\wedge e$ 1
4. $q \wedge r$ $\wedge i$ 3,2

When clear we omit subscripts e_1, e_2 and just write e .